PROSPECTIVE ED PILOT COHORT · ABSTRACT SUBMISSION · SYNTHETIC DATA

Point-of-Care Capillary Toe-to-Finger Lactate Gradient During Early Emergency Department Sepsis Resuscitation

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Institutional affiliation not specified in source abstract (Synthetic Pilot Submission)

Clinical Context

Macrohemodynamic stabilization during early septic shock resuscitation can mask persistent peripheral microcirculatory hypoperfusion.

- **Objective:** Test whether bedside capillary toe-to-finger lactate gradient (ΔLac) at 2 hours predicts 72-hour vasopressor dependency.

Pilot Study Design (N = 164)

- **Setting & Cohort:** Prospective single-ED observational pilot study of N = 164 adults with suspected septic shock.
- **Measurement:** Paired fingertip and hallux point-of-care capillary lactate sampled at baseline (0 h) and 2 hours post-crystalloid resuscitation.
- **Pre-specified Exposure Threshold:** 2-hour ΔLac ≥ 1.2 mmol/L (n = 58, 35.4%) vs. ΔLac < 1.2 mmol/L (n = 106, 64.6%).

Source Scope & Transparency Note

Designed strictly from the accepted pilot abstract. Multivariable adjusted models, secondary organ-dysfunction endpoints, and formal limitation statements were not reported in the abstract and were intentionally omitted without fabrication.

Principal Pilot Finding

RR 2.86 (95% CI 1.98–4.13; p = 0.0001)

Patients with persistent 2-hour capillary ΔLac ≥ 1.2 mmol/L experienced nearly threefold higher 72-hour vasopressor dependency (62.1% vs. 21.7%).

72-Hour Vasopressor Dependency by 2-Hour ΔLac

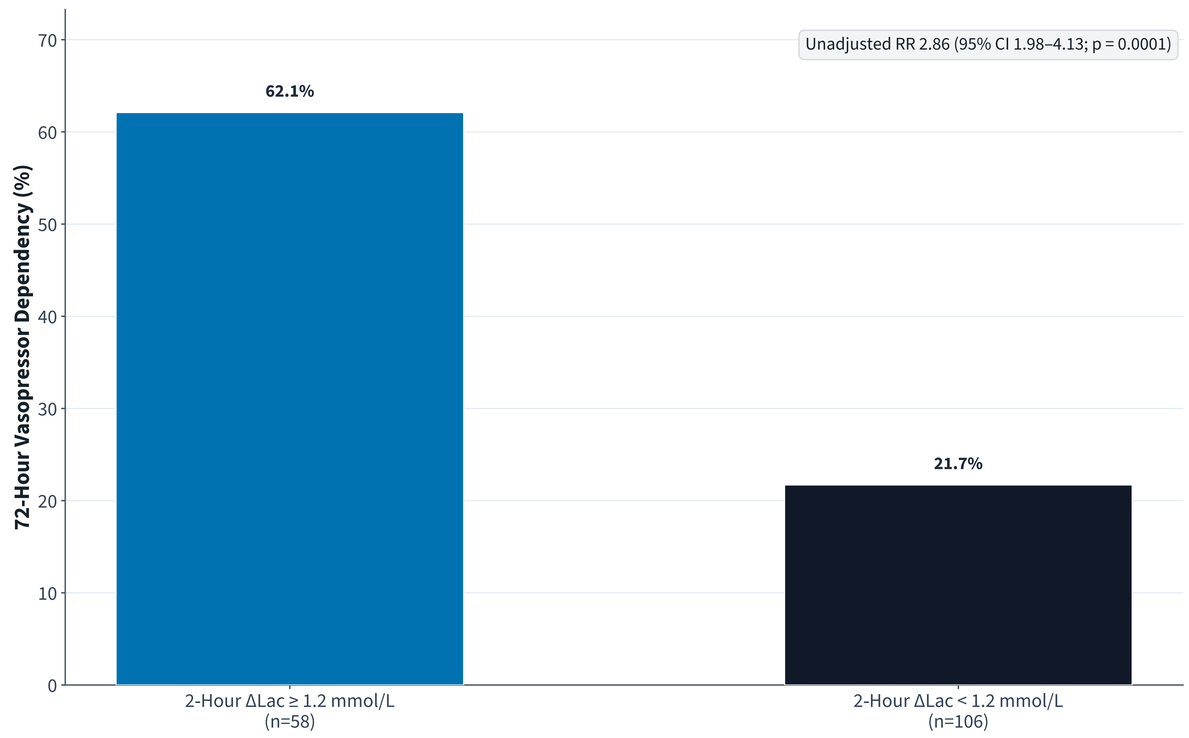


Figure 1. Observed 72-hour vasopressor dependency across 2-hour capillary lactate gradient strata (N = 164; exact binomial proportions).

Conclusions & Next Steps

Bedside 2-hour capillary toe-to-finger lactate gradient rapidly identifies occult microcirculatory failure during initial ED sepsis resuscitation and warrants prospective multicenter validation.

SYNTHETIC PILOT ABSTRACT DISCLOSURE: Created strictly to validate missing-data handling without hallucination.